Zuodong, WANG
王作栋

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wangzuodong1997.github.io

RESEARCH INTERESTS

Numerical Analysis

Fluid dynamics, semi-linear parabolic problems, PDEs with stiff source term, structure-preserving schemes, a posteriori error analysis

Programming

C++, Python, Matlab, Julia

ACADEMIC APPOINTMENTS

Postdoctoral Researcher (2025-present)

Eastern institute of technology, Ningbo link

- Project: Efficient high-order structure-preserving schemes for evolution problems
- Advisor: Jie Shen

EDUCATION

PhD (2025)

CERMICS and SERENA team,
Ecole nationale des Ponts et Chaussées, Institut Polytechnique de Paris & Inria, Paris

- Thesis title: Efficient numerical schemes for evolution equations with singularities and shocks
- Advisors: Zhaonan Dong, Alexandre Ern

Master (2022)

Ecole nationale des Ponts et Chaussées, ParisTech
& Sorbonne Université

- Applied Mathematics: Modeling, Analysis and Simulation
- Grade: 16.10/20 (top class)

Engineer's degree (2021-2022)

Ecole nationale des Ponts et Chaussées, department IMI

Bachelor (2020)

Beijing University of Technology

- Mathematics and Applied Mathematics
- GPA: 3.77/4.0

LANGUAGES

Chinese (Mandarin): Native,
English: Fluent, French: Fluent, Japanese: Duolingo

HOBBIES

ACG(Anime, Comics, Games)

CONFERENCES

24th annual Meeting of China Society for Industrial and Applied Mathematics
Lanzhou, China, 2026 – *contributed talk*

University of Nottingham Ningbo China Conference on Applied Mathematics and Mathematical Biology
Ningbo, China, 2026 – *invited talk*

11th International Conference on Computational Methods in Applied Mathematics
Vienna, Austria, 2026 – *invited talk*

Tsinghua Sanya International Mathematics Forum on Advanced numerical schemes for multiphysics problems
Sanya, China, 2025 – *invited talk*

30th Biennial Conference on Numerical Analysis
Glasgow, UK, 2025 – *contributed talk*

International Conference on Applied Mathematics
Hongkong, China, 2024 – *contributed talk*

Finite Element Rodeo
Houston, USA, 2024 – *talk in mini-symposium*

The 6th SIAM Texas-Louisiana Sectional Meeting
Lafayette, USA, 2023 – *talk in mini-symposium*

TEACHING

Numerical analysis (French)
2025 – *Paris Dauphine University*

Explorations in Mathematics (English)
2024 – *Texas A&M university*

AWARDS

CSC Scholarship
Paris, 2020 – 2022

Outstanding Graduate
Beijing University of Technology, 2020

Model Student of Academic Records
Beijing University of Technology, 2016 – 2020

EXPERIENCE

Visiting scholar

Texas A&M university
College station, USA, September 2023 – May 2024

- Project: Design, analyse and testing Invariant-domain-preserving schemes for hyperbolic problems

Internship (stage court)

Inria Centre de Paris
Paris, France, June – September 2021

- Project: Hybrid high-order method for elliptic PDEs on curved and complicated domains

PUBLICATIONS

1. **Two nodally bound-preserving and mass-conservative discontinuous Galerkin methods for reaction-diffusion problems**
   2026, Jie Shen, Zuodong Wang – *Submitted*

2. **Pressure-robust hp-a posteriori error estimates of H(div)-conforming discontinuous Galerkin methods for the Stokes equations**
   2026, Zhaonan Dong, Zuodong Wang, Lina Zhao – *Submitted* link

3. **A high-order nodally bound-preserving and mass-conservative method for linear fourth-order elliptic problems and its applications to nonlinear parabolic equations**
   2026, Jie Shen, Zuodong Wang – *Submitted* link

4. **A posteriori error analysis and adaptivity of a space-time finite element method for the wave equation in second order formulation**
   2026, Zhaonan Dong, Emmanuil H Georgoulis, Lorenzo Mascotto, Zuodong Wang – *Numerische Mathematik* link

5. **A priori and a posteriori error estimates of a $\mathcal{C}^0$-in-time method for the wave equation in second order formulation**
   2026, Zhaonan Dong, Lorenzo Mascotto, Zuodong Wang – *Numerische Mathematik* link

6. **A bound-preserving scheme for the Allen-Cahn equation**
   2025, Zhaonan Dong, Alexandre Ern, Zuodong Wang – *Computers & Mathematics with Applications* link

7. **Mass conservative limiting and applications to the approximation of the steady-state radiation transport equations**
   2025, Jean-Luc Guermond, Zuodong Wang – *Journal of Computational Physics* link

8. **Asymptotic and invariant-domain preserving schemes for scalar conservation equations with stiff source terms and multiple equilibrium points**
   2024, Alexandre Ern, Jean-Luc Guermond, Zuodong Wang – *Journal of Scientific Computing* link

9. **Hybrid high-order methods for elliptic PDEs on curved and complicated domains**
   2022, Zhaonan, Dong, Zuodong Wang – *13th ICOSAHOM 2020/2021* link

10. **The Proof of 2n Circle Arrangement Conjecture**
    2017, Zuodong Wang – *Pure Mathematics* link